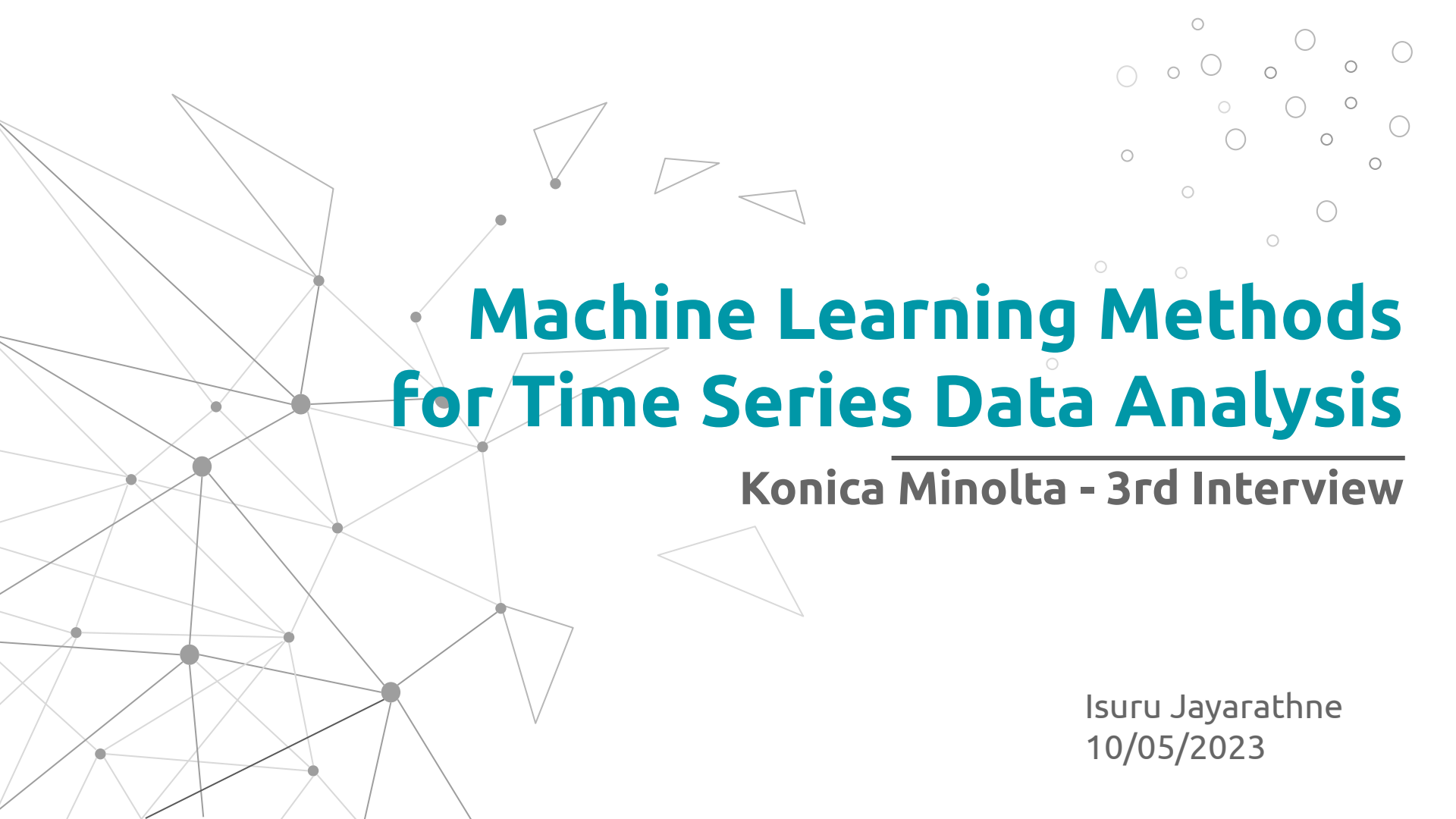




Machine Learning Methods for Time Series Data Analysis

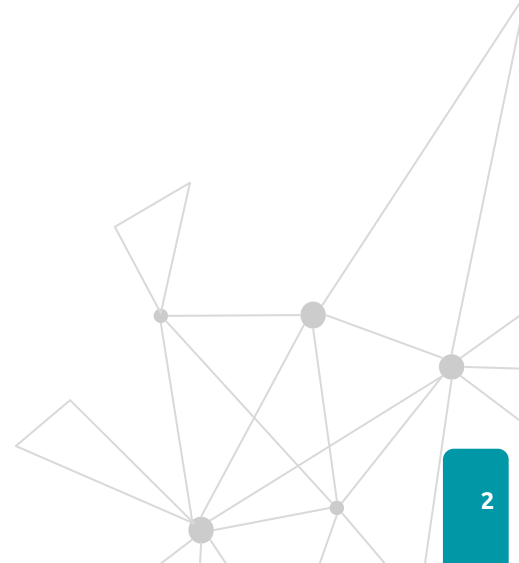
Konica Minolta - 3rd Interview

Isuru Jayarathne
10/05/2023



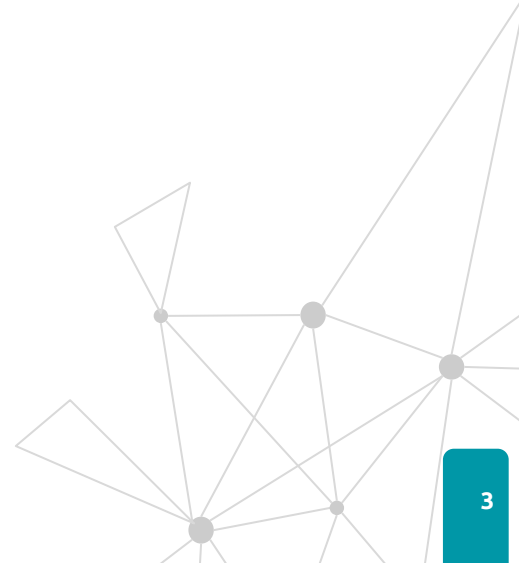
Contents

- Main Categories
- Feedback Networks
- Encoder, Decoder Architecture
- HiPPO framework
- Linear State-Space Layers
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- References



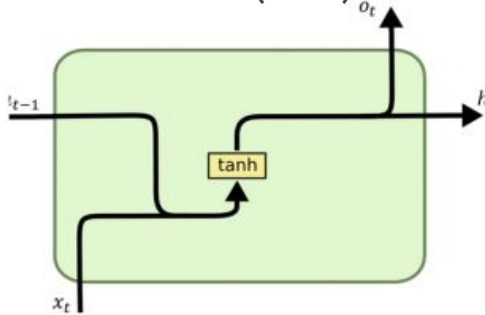
Main Categories

- Feedback Networks (or RNN)
 - RNN, LSTM, GRU, LMU, HiPPO-RNN
- Encoder, Decoder based architecture (Attention mechanism)
 - Transformers, BERT, GPT
- Temporal Convolutional Networks (TCN)
 - GTCN, DTCN, STCN, RTCN, ATCN
- Ordinary Differential Equation Networks (ODE-Net)
 - NDE, Residual ODE-Net
- Other
 - LSSL

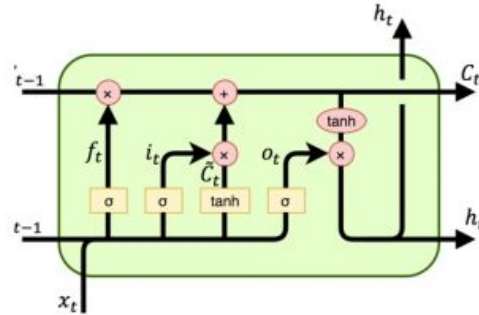


Feedback Networks

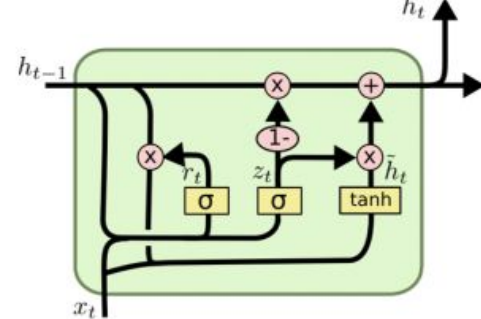
Recurrent Neural Network (RNN)



Long Short Term Memory (LSTM)



Gated Recurrent Unit (GRU)



Source: <http://dprogrammer.org/rnn-lstm-gru>

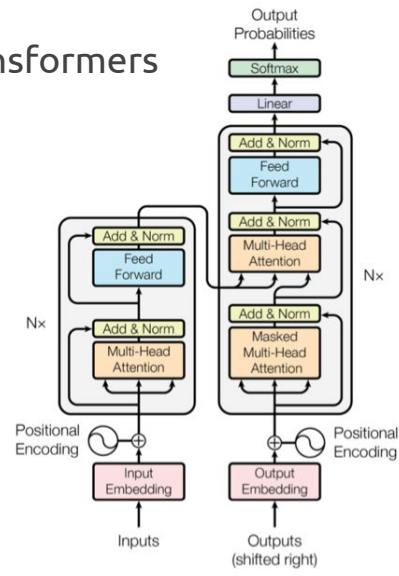
- Vanishing/exploding gradient problem

- No vanishing gradient
- Computationally inefficient

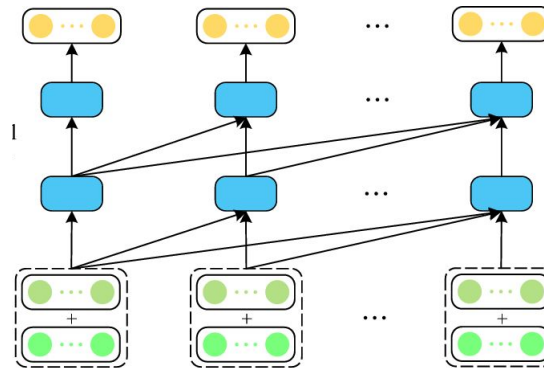
- Simpler and computationally efficient architecture

Encoder, Decoder Architecture

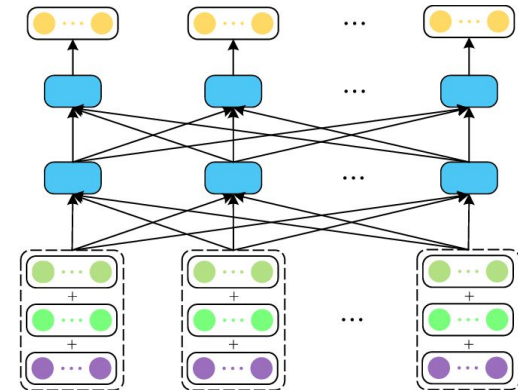
Transformers



Generative
Pre-trained
Transformers (GPT)



Bidirectional Encoder
Representations from
Transformers (BERT)



Source: 10.1007/s13042-020-01069-8

- Widely use in NLP domain (can be used for time-series data)
- Use encoders and decoders to learn the context
- Capable of dealing with long sequences



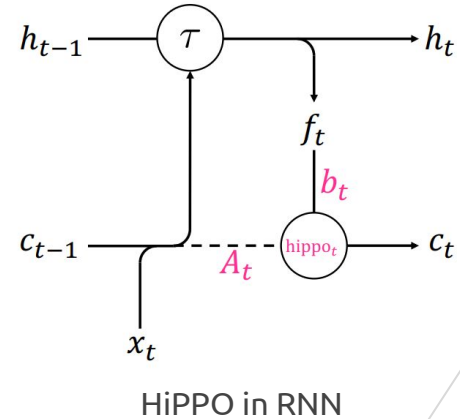
HiPPO framework

- Stand for Higher Order Polynomial Projection Operator
- A solution for online function approximation problem
- Supervised learning method for regression tasks.
- Generate polynomial features and project them into lower-dimensional space using projection operator
- Predictions are made by feeding projected features to regularized linear model



HiPPO framework cont.

- Can be used with different types of polynomial basis such as Fourier, Chebyshev, Legendre.
- HiPPO can also be combined with RNNs to model non-linear time series data.
- HiPPO-LegS which uses Legendre bases has outperformed most of SotA models (LSTM, LMU, Transformer, TCN)
- Hyperparameter-free HiPPO-LegS also has a potential to remember all history



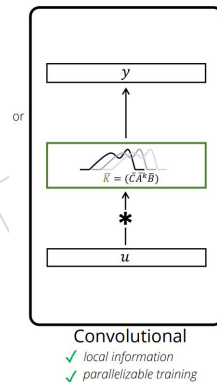
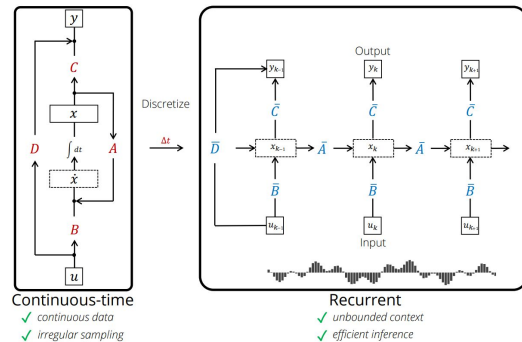
Linear State-Space Layers

- Based on state-space equations in control systems
- Strengths over other methods:
 - Stateful model for sequential data (RNNs +)
 - Not slow to train and do not suffer from vanishing gradient (RNNs -)
 - Encode local context, fast, parallelizable training (CNNs +)
 - No limitation on context length (CNNs -)
 - Address continuous-time and long-term dependencies (NDEs +)
 - Efficient compared to NDEs (NDEs -)

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

Linear State-Space Layers cont.

- Can be used as RNN, CNN, or NDE
- LSSL maps a sequence of inputs to outputs by simulating a linear continuous-time state-space representation
- By adding trainable subset of structured matrices LSSL has been enhanced with long-range memory capability.
- Also, a deep neural network can be made by stacking LSSL layers.



Structured State Space (S4)

- Based on state-space model (SSM)
- Can address long-range dependencies in sequential data
- Introduced to address computational bottleneck problem in LSSL architecture
- This issue has been overcome by parameterized the state space in 3 stages
 - Compute spectrum by evaluating its truncated generating function and then apply iFFT
 - Apply Woodbury identity to correct the low rank term
 - Cauchy kernel is used to do the final computation



Structured State Space (S4) cont.

- This technique can be used with any matrix that can be decomposed as Normal Plus Low-Rank (NPLR) matrices
- S4 can be used as deep learning method with less computational complexity compared to recurrent and attention models
- In addition, S4 has outperformed most SotA techniques for sequential data with benchmark datasets
- In conclusion, S4 can be used as a general deep learning method for sequential data such as images, audio, text, and time-series

References

- Gu, A., Dao, T., Ermon, S., Rudra, A., & Ré, C. (2020). Hippo: Recurrent memory with optimal polynomial projections. Advances in neural information processing systems, 33, 1474-1487.
- Gu, A., Goel, K., & Ré, C. (2021). Efficiently modeling long sequences with structured state spaces. arXiv preprint arXiv:2111.00396.
- Gu, A., Goel, K., & Ré, C. (2021). Efficiently modeling long sequences with structured state spaces. arXiv preprint arXiv:2111.00396.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. Advances in neural information processing systems.

Videos:

- <https://crossminds.ai/video/hippo-recurrent-memory-with-optimal-polynomial-projections-606fd7f743a7f2f827bf91d/>
- https://www.youtube.com/watch?v=luCBXCErkCs&ab_channel=StanfordMedAI
- https://www.youtube.com/watch?v=EvQ3ncuriCM&ab_channel=StanfordMLSysSeminars
- https://www.youtube.com/watch?v=6MAF6upe13E&ab_channel=SanFranciscoBayACM



Thank you
